

Biostatistics: Definition, Statistical Methods, Basic Principles, Variables and Applications

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What is Biostatistics?

Biostatistics is the application of statistical principles and methods to biological, medical, public health, agricultural, and biotechnology-related data.

It helps researchers to:

- Collect biological data systematically.
- Summarize large datasets meaningfully.
- Compare treatment groups.
- Test scientific hypotheses.
- Predict biological outcomes.
- Make evidence-based decisions.

Why Biostatistics is Important in Biotechnology

Biotechnology generates large and complex datasets such as:

- Enzyme activity data
- Microbial growth data
- Gene expression values
- Clinical trial observations
- Plant tissue culture response
- Drug dose–response measurements
- Bioinformatics and omics datasets

Without statistics, biological data remain observations. With statistics, they become evidence.

Example Dataset: Plant Growth Experiment

Research question: Does biofertilizer improve plant height?

Plant ID	Group	Height (cm)	Leaves
1	Control	18	6
2	Control	20	7
3	Control	19	6
4	Biofertilizer	27	9
5	Biofertilizer	29	10
6	Biofertilizer	26	9

Statistical use: Compare mean height between control and treated plants.

Types of Statistical Methods

① Descriptive Statistics

- Summarizes data using mean, median, mode, range, variance, SD.

② Inferential Statistics

- Draws conclusions about a population using sample data.

③ Experimental Design

- Plans experiments using randomization, replication, and controls.

④ Predictive Statistics

- Uses regression, classification, and models to predict outcomes.

Descriptive Statistics

Descriptive statistics help describe the basic features of data.

Common Measures

- **Mean:** Average value
- **Median:** Middle value
- **Mode:** Most frequent value
- **Range:** Maximum – Minimum
- **Standard Deviation:** Spread around the mean

Example: Heights of treated plants: 27, 29, 26 cm

$$\text{Mean} = \frac{27 + 29 + 26}{3} = 27.33 \text{ cm}$$

Inferential Statistics

Inferential statistics help make conclusions beyond the collected data.

- Is the difference between two groups real or due to chance?
- Is a drug effective?
- Is there an association between exposure and disease?
- Can sample results represent the population?

Common Inferential Tests

- Student's t-test
- Chi-square test
- ANOVA
- Correlation
- Regression

Basic Principle 1: Data Must Answer a Question

Every statistical analysis starts with a clear research question.

Example

Poor question: Study plants.

Good question: Does biofertilizer increase mean plant height compared with untreated control plants?

A good question defines:

- Population
- Variable
- Comparison group
- Expected outcome

Basic Principle 2: Sampling

A **sample** is a subset selected from a larger population.

- Population: All final-year biotechnology students
- Sample: 50 students selected for a survey

Good Sampling Should Be

- Representative
- Unbiased
- Adequate in sample size
- Suitable for the study objective

Basic Principle 3: Variation is Natural

Biological systems show natural variability.

- Human blood glucose differs between individuals.
- Enzyme activity varies between samples.
- Plant growth varies due to genetics and environment.
- Microbial growth varies with pH, temperature, and nutrients.

Statistics helps distinguish natural variation from meaningful biological effects.

Basic Principle 4: Control, Randomization and Replication

Control

- Baseline comparison
- Example: untreated plants

Randomization

- Avoids selection bias
- Assigns samples randomly

Replication

- Repeated observations
- Improves reliability

Key Message

A biological experiment without control, randomization, and replication may produce misleading conclusions.

Variables in Biostatistics

A **variable** is any characteristic that can vary among individuals, samples, or experimental units.

Examples in Biotechnology

- Height of plants
- Enzyme activity
- Blood glucose level
- Type of treatment
- Gene expression level
- Presence or absence of infection

Classification of Variables

Variable Type	Description	Example
Qualitative	Categories	Blood group
Quantitative	Numerical values	Plant height
Discrete	Countable numbers	Number of colonies
Continuous	Measurable values	Body temperature
Independent	Cause/predictor	Fertilizer dose
Dependent	Effect/outcome	Plant growth
Confounding	Hidden influence	Soil quality

Qualitative Variables

Qualitative variables describe categories or groups.

- Sex: Male / Female
- Blood group: A / B / AB / O
- Disease status: Positive / Negative
- Treatment group: Control / Drug / Vaccine
- Bacterial colony type: Smooth / Rough

Common analysis:

- Frequency table
- Percentage
- Bar diagram
- Chi-square test

Quantitative Variables

Quantitative variables are numerical measurements.

- Weight of tissue culture callus
- Optical density of bacterial culture
- Blood pressure
- Enzyme activity
- Protein concentration
- DNA yield

Common analysis:

- Mean
- Standard deviation
- t-test
- ANOVA
- Correlation and regression

Measurement Scales

Scale	Meaning	Biological Example
Nominal	Name/category only	Blood group
Ordinal	Ordered category	Disease severity
Interval	Equal interval, no true zero	Temperature in Celsius
Ratio	Equal interval with true zero	Weight, height, enzyme activity

Important: The measurement scale determines the correct statistical test.

Measurement: Accuracy, Precision, Reliability

Accuracy

Closeness of measured value to the true value.

Precision

Closeness of repeated measurements to each other.

Reliability

Consistency of measurement over repeated trials.

Example

If a micropipette repeatedly delivers 95 μL instead of 100 μL , it is precise but not accurate.

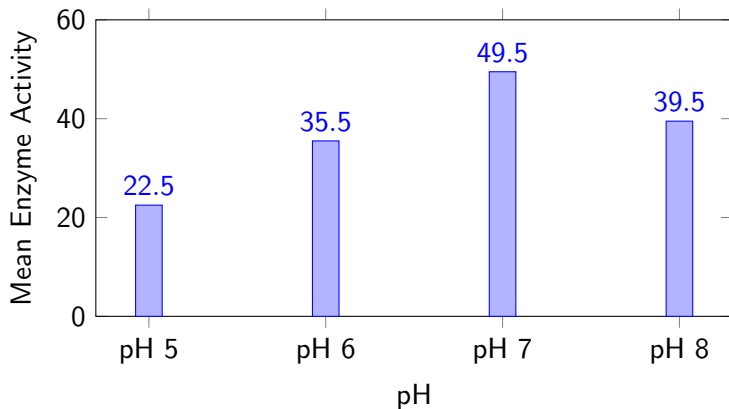
Dataset Example: Enzyme Activity

Sample	pH 5	pH 6	pH 7	pH 8
1	22	35	48	39
2	24	37	51	41
3	21	36	49	38
4	23	34	50	40

Research question: At which pH is enzyme activity maximum?

Possible method: One-way ANOVA.

Visualizing Enzyme Activity



Interpretation: Enzyme activity appears highest at pH 7.

Functions of Statistics

Statistics helps in:

- ① **Data collection** – planning surveys and experiments.
- ② **Data organization** – tables and charts.
- ③ **Data summarization** – mean, median, SD.
- ④ **Comparison** – treatment vs control.
- ⑤ **Prediction** – disease risk, growth rate.
- ⑥ **Decision making** – accepting or rejecting hypotheses.
- ⑦ **Scientific communication** – presenting evidence clearly.

Uses of Statistics in Biotechnology

- Comparing microbial growth under different media.
- Estimating LD_{50} or IC_{50} values.
- Evaluating vaccine efficacy.
- Analyzing ELISA absorbance values.
- Studying gene expression differences.
- Optimizing fermentation conditions.
- Interpreting clinical trial data.
- Validating diagnostic test performance.

Example: Diagnostic Test Dataset

	Disease Present	Disease Absent
Test Positive	45	10
Test Negative	5	40

$$\text{Sensitivity} = \frac{45}{45 + 5} \times 100 = 90\%$$

$$\text{Specificity} = \frac{40}{40 + 10} \times 100 = 80\%$$

Use: Statistics helps evaluate diagnostic test accuracy.

Limitations of Statistics

Statistics is powerful, but it has limitations.

- It cannot correct poor experimental design.
- It cannot prove causation from correlation alone.
- Results may be misleading if sample size is small.
- Biased data produce biased conclusions.
- Statistical significance may not always mean biological importance.
- Missing data can affect interpretation.
- Incorrect test selection gives invalid results.

Common Mistakes in Biostatistics

Avoid These Errors

- Using mean for highly skewed data without checking distribution.
- Ignoring outliers.
- Treating ordinal data as continuous data.
- Using t-test for more than two groups instead of ANOVA.
- Reporting only p-value without effect size.
- Drawing causal conclusions from observational data.

Choosing the Right Statistical Method

Research Question	Data Type	Method
Mean of one group	Quantitative	One-sample t-test
Two group comparison	Quantitative	Independent t-test
More than two groups	Quantitative	ANOVA
Association between categories	Qualitative	Chi-square test
Relationship between variables	Quantitative	Correlation
Prediction of outcome	Quantitative	Regression

Mini Case Study

Experiment: A researcher tests whether three media improve bacterial growth.

Replicate	Medium A	Medium B	Medium C
1	0.42	0.58	0.76
2	0.45	0.61	0.79
3	0.43	0.59	0.81
4	0.46	0.63	0.78

Variable: Optical density at 600 nm

Suitable test: One-way ANOVA

Interpretation of Mini Case Study

- Medium C shows the highest OD values.
- Medium A shows the lowest OD values.
- ANOVA can test whether the differences are statistically significant.
- Post-hoc tests can identify which media differ from each other.

Biological Meaning

If Medium C significantly increases OD, it may be better for bacterial biomass production.

Statistics and Scientific Decision Making

Statistical analysis supports scientific decisions by answering:

- Is there a measurable difference?
- Is the difference likely due to treatment?
- Is the result statistically significant?
- Is the result biologically meaningful?
- Can the result be reproduced?
- Can the conclusion be generalized?

Key Takeaways

- Biostatistics converts biological observations into scientific evidence.
- Statistical methods include descriptive, inferential, experimental, and predictive approaches.
- Good statistics begins with good study design.
- Variables may be qualitative or quantitative.
- Measurement scale determines the correct statistical test.
- Statistics has limitations and must be interpreted carefully.
- Biotechnology research depends on statistical reasoning for valid conclusions.

Student Activity

Using the plant growth dataset:

- 1 Identify the independent variable.
- 2 Identify the dependent variable.
- 3 Calculate mean height for each group.
- 4 Decide which statistical test can be used.
- 5 Write one biological conclusion.

Expected answer: Independent variable = treatment group; dependent variable = plant height.

Recommended Reading

- Daniel, W. W. *Biostatistics: A Foundation for Analysis in the Health Sciences*.
- Zar, J. H. *Biostatistical Analysis*.
- Rosner, B. *Fundamentals of Biostatistics*.
- Motulsky, H. *Intuitive Biostatistics*.

Thank You

Questions and Discussion